



# Data Warehousing Lab: Dashboarding with Tableau

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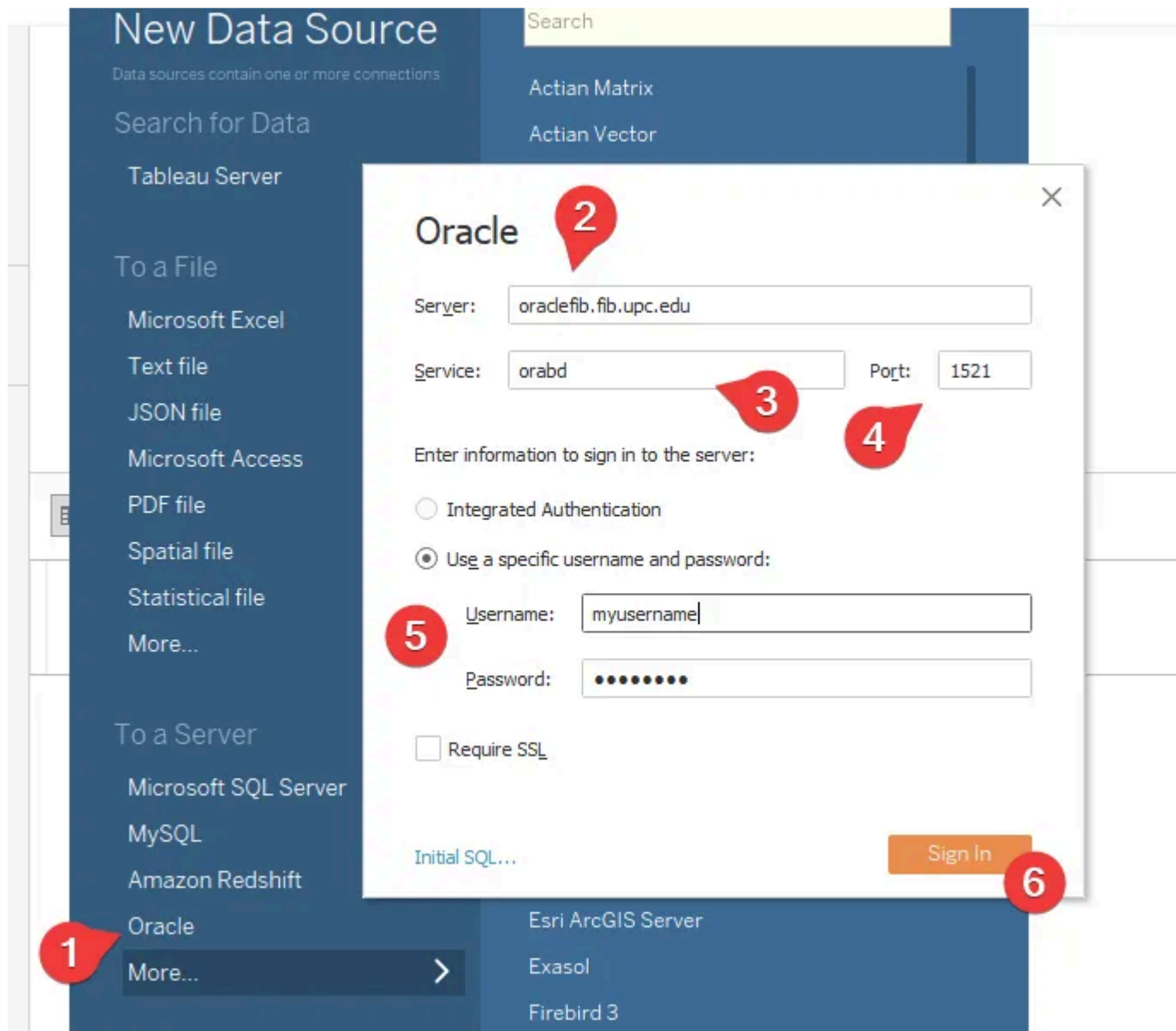
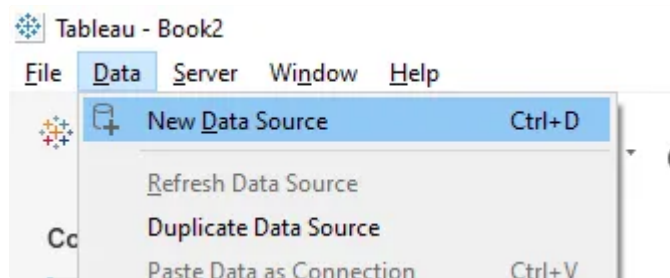
## Wrapping up

## Introduction

This is a quick overview of the things you will need to handle in the project.

## Connecting to the target DW (Oracle database)

First, we will see how to connect to our target DW, previously created in Oracle and with data loaded with the data generation script.



- You might run into an error at this moment: **Error connecting to the DB** 💣

## Loading the data model

After connecting to the DB, on the left we will have the tables and views created in our DW. These tables/views are either factual or dimensional.



The concepts appearing in the left sidebar are all of the tables and views (materialized or virtual, if any) present in the database you are connecting to. Here, for example, `V_AIRCRAFT_UTILIZATION_A_M` is a view over some of the tables in the DB.

We now start creating the data model (multidimensional cube) that will be used in the dashboards.

To do so, we need to drag the blocks (tables or views) into the data model canvas in the top right region (as shown below). We first create a fact concept. It can be created by a single factual table or view or even combining multiple facts (i.e., doing drill across). To combine multiple facts, open the details of the first dragged fact and then drag another one in the same canvas (as shown below). After the second fact is set, you can configure the drill across join by clicking in the Venn diagram between the facts.

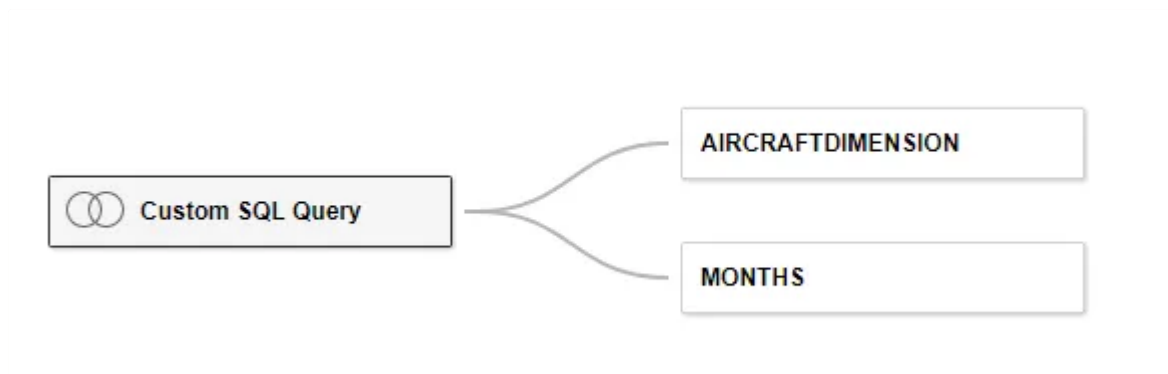
Select **inner join** and pick the attributes from each query that you wish to perform the join on. In this case, we join on the **aircraft** and **month** attributes.

The screenshot shows the 'Join' dialog box in Tableau. At the top, it says 'Custom SQL Query is made of 2 tables. ⓘ'. Below this, two data sources are connected by a Venn diagram icon. The 'Join' dialog box is open, showing four options: Inner, Left, Right, and Full Outer. The 'Inner' option is selected. Below the join type, there is a table showing the join conditions:

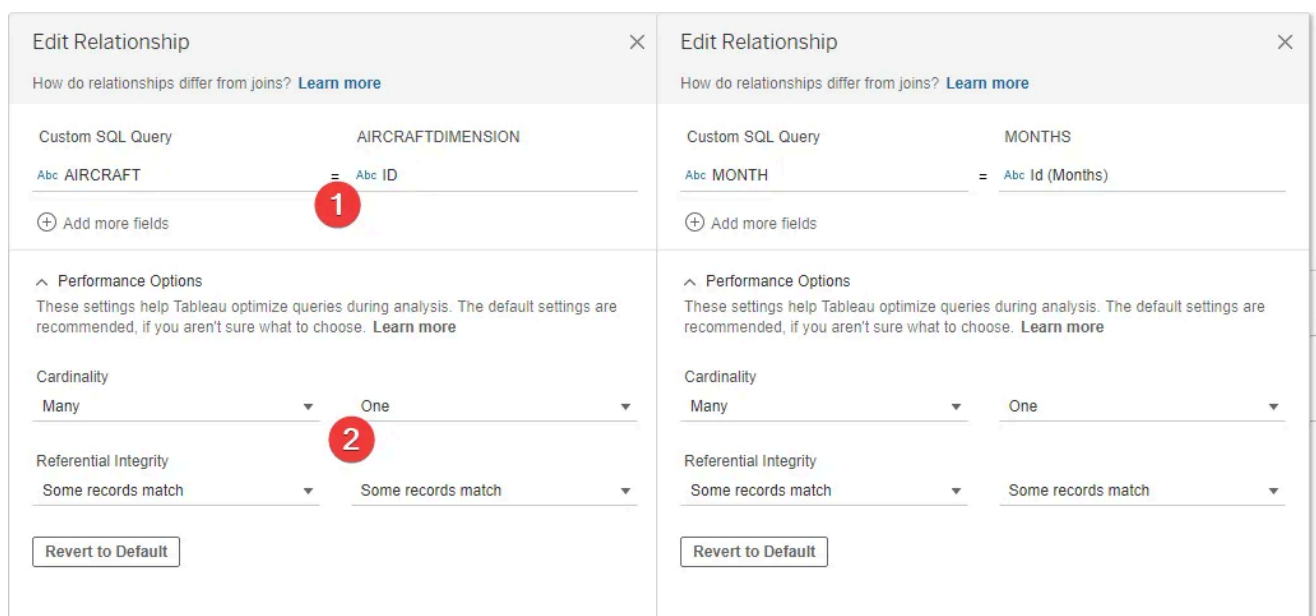
| Data Source                |   | Custom SQL Query1     |
|----------------------------|---|-----------------------|
| MONTH                      | = | MONTH (Custom SQL ... |
| AIRCRAFT                   | = | AIRCRAFT (Custom S... |
| <i>Add new join clause</i> |   |                       |

At the bottom of the dialog, there is a list of fields from both data sources. The fields from 'Custom SQL Query' are AIRCRAFT, MONTH, FH, FC, ADIS, ADOSS, and ADOSU. The fields from 'Custom SQL Query1' are MONTH, AIRCRAFT, and ADOSU.

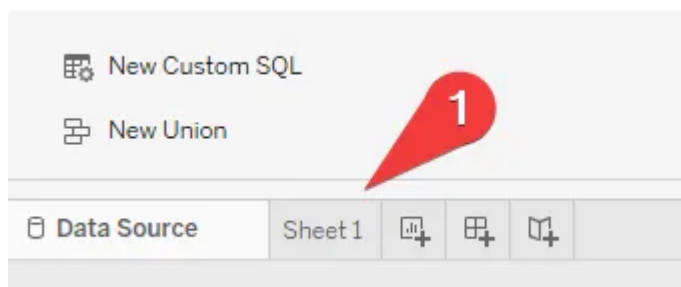
After setting up the fact concept (that can consist of drill across between two tables or views as seen in the previous example), now, we include the dimensions (e.g., **AIRCRAFTDIMENSION** and **MONTHS** to the diagram), by dragging the corresponding tables them from the left side menu into the main canvas of the data model.



You will have to set the join attributes manually **1**, by clicking in the lines that link factual and dimensional blocks. Remember also that the cardinality of these relationships should be set to **many to one** **2** in order to satisfy multidimensional integrity constraints.



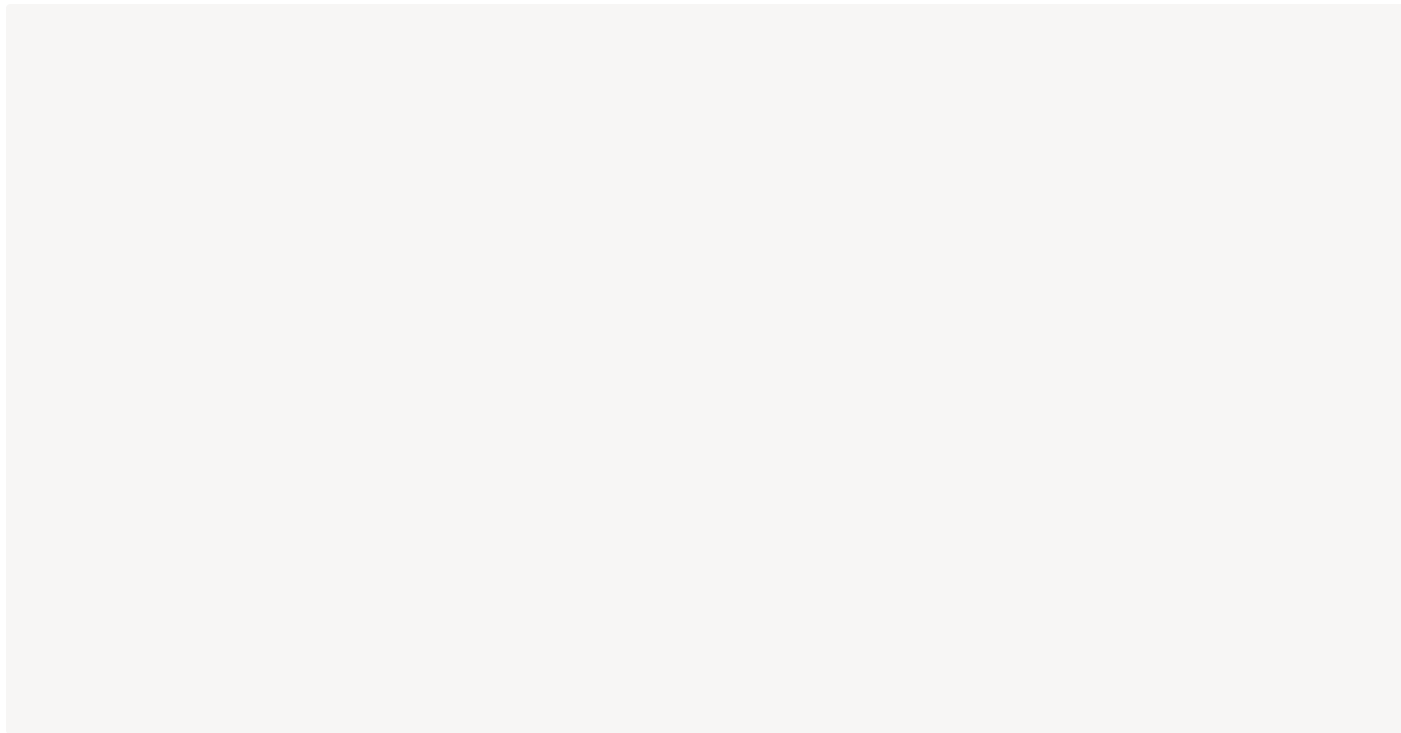
Once the multidimensional data model is set, we are now ready to start creating charts for our dashboard. Click in the *Sheet1* tab below



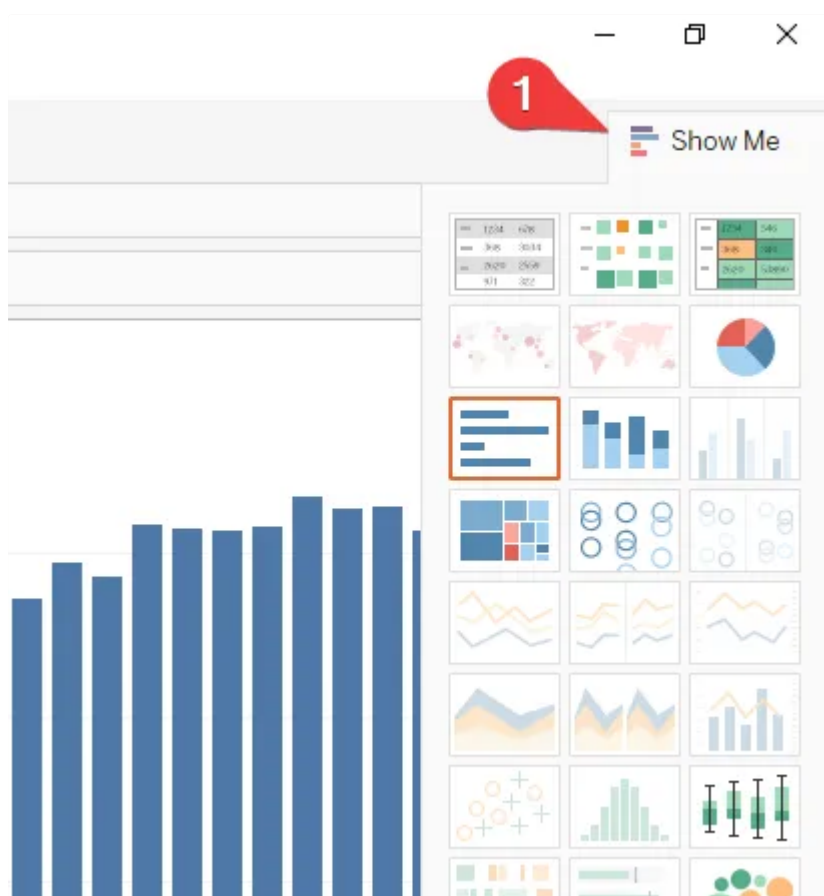
## A basic plot of the delay count per month

We will start with a basic plot, by posing a question: *What is the delay count per month?*

We start off by dragging the entity **DY** from the left sidebar, to the *rows* area, and the **Id(Month)** to the *columns* area (as shown below).

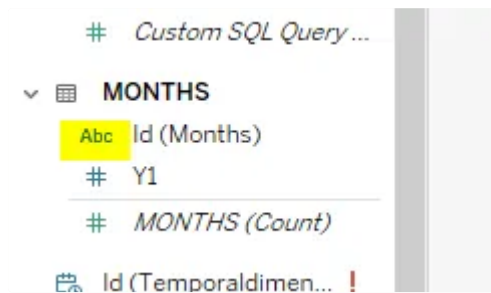


You can change the chart type depending on the data you provide. Check your options by clicking in the *Show me* button.

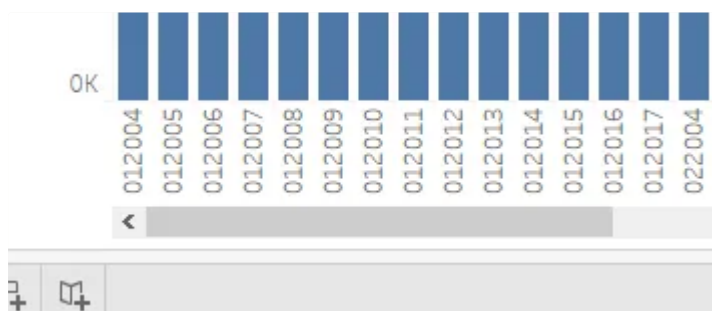


## On parsing strings to months

You will note that the `Id(Month)` attribute is of `string` type. How do we know that?



By default these months do not have the order we would like them to have. We will have to convert them into proper `date` values.



There are many ways you can achieve this, depending on what data you have at the moment. You can read more from the documentation

- [https://help.tableau.com/current/pro/desktop/en-us/data\\_dateparse.htm](https://help.tableau.com/current/pro/desktop/en-us/data_dateparse.htm), for how to parse `date` objects
- [https://help.tableau.com/current/pro/desktop/en-us/functions\\_functions\\_string.htm](https://help.tableau.com/current/pro/desktop/en-us/functions_functions_string.htm), for how to manipulate `string` objects.

We will parse the strings into proper dates by using `DATEPARSE`, and by passing the string formatted as `"MM/yyyy"`.

**!** Here, we are considering dates in the format `dd/MM/yyyy`. This depends on your locale and it may vary depending on your system. You can change the locale Tableau is using for a given workbook, [read more here](#).

### 1. Create a dimension from `Id (Months)`

`DATEPARSE("MM/yyyy", LEFT([Id (Months)],2) + "/" + RIGHT([Id (Months)], 5))`. Make sure you understand the different parts involved in this transformation. For example, function `LEFT` is extracting from the string in the first parameter the leftmost substring of the length defined by a second parameter (e.g., 2). Similarly, function `RIGHT` is extracting the rightmost substring.



The `DATEPARSE` function converts a string to a date object, and you can manipulate it with `DATE` functions. Check [https://help.tableau.com/current/pro/desktop/en-us/functions\\_functions\\_date.htm](https://help.tableau.com/current/pro/desktop/en-us/functions_functions_date.htm) for more info.

After you are done with this, you should have converted the strings, e.g.

`012020` to `1/01/2020`, and then to a date object. Alternatively, you can create a new field directly from the attribute

The screenshot shows the Tableau interface with a context menu open over a table named 'MONTHS'. The menu options include 'Add to Sheet', 'Show Filter', 'Duplicate', 'Rename', 'Hide', 'Aliases...', 'Create', 'Transform', 'Convert to Measure', 'Change Data Type', 'Geographic Role', 'Default Properties', 'Group by', 'Folders', 'Hierarchy', 'Replace References...', and 'Describe...'. The 'Create' option is selected, and a sub-menu is open showing 'Calculated Field...', 'Group...', 'Set...', and 'Parameter...'. The 'Calculated Field...' option is highlighted.

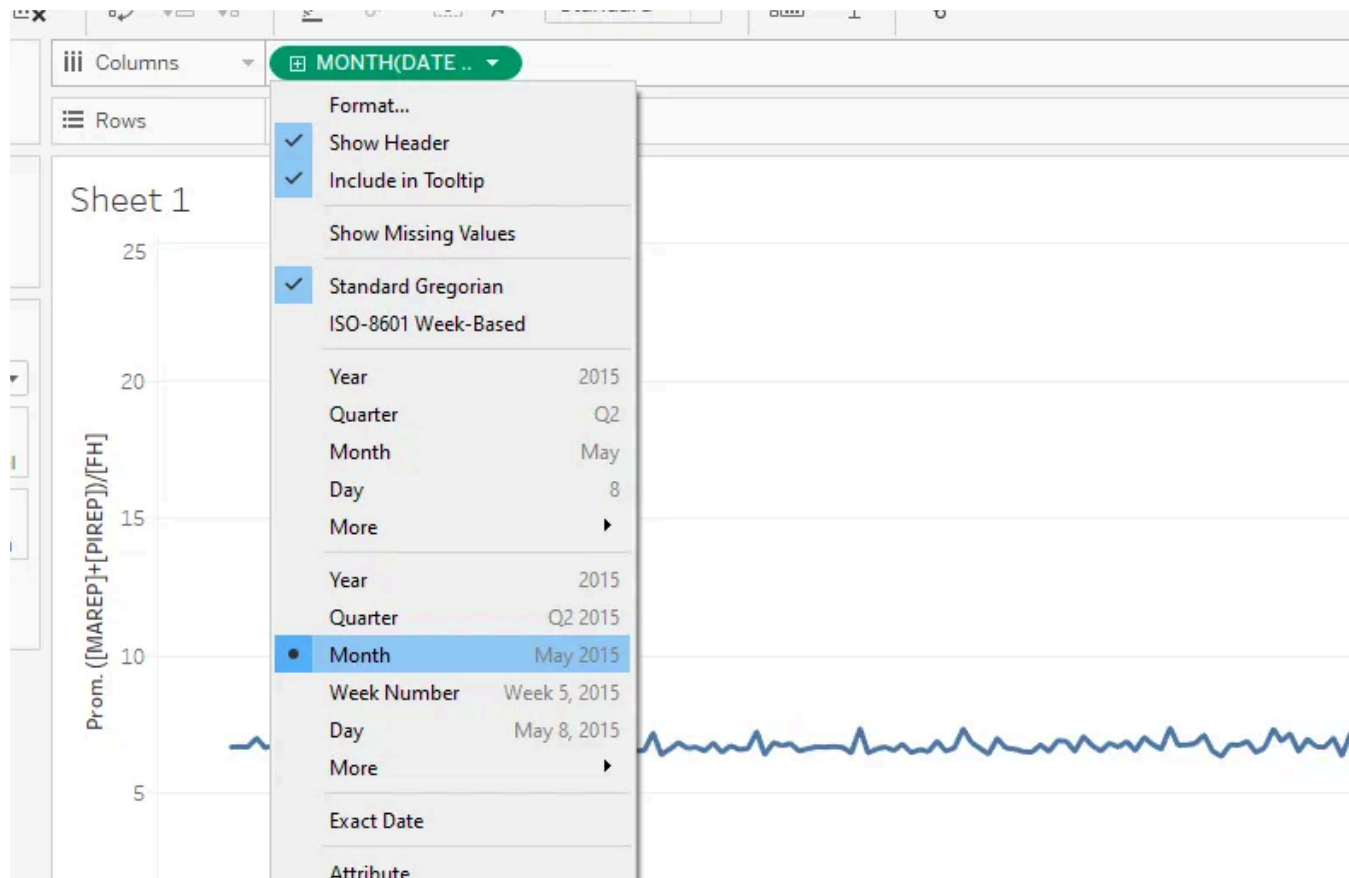
The background shows the 'Data' pane with a search bar and a list of tables. The 'Columns' and 'Rows' shelves are visible on the right. A bar chart is displayed on the right side of the view.

Below the main interface, a validation dialog box is open. It contains the text 'parsed\_datetime' in the input field and the following SQL query in the text area:

```
DATEPARSE("MM/yyyy", LEFT([Id (Months)], 2) + "/" + RIGHT([Id
```

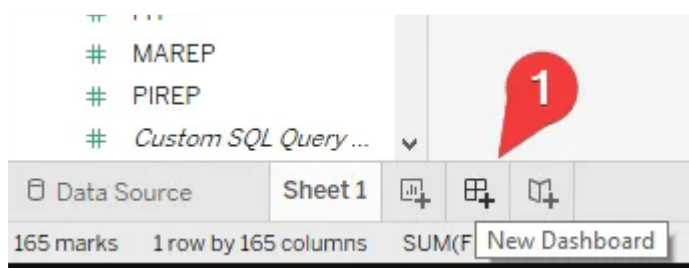
The dialog box also displays the message 'The calculation is valid.' and has 'Apply' and 'OK' buttons.

- You can now select the *month* from the dropdown, as a continue value (in green). If you were to select it as a discrete measure, the aggregation would happen by unique months instead of a continuous timeline. Check it by yourself.

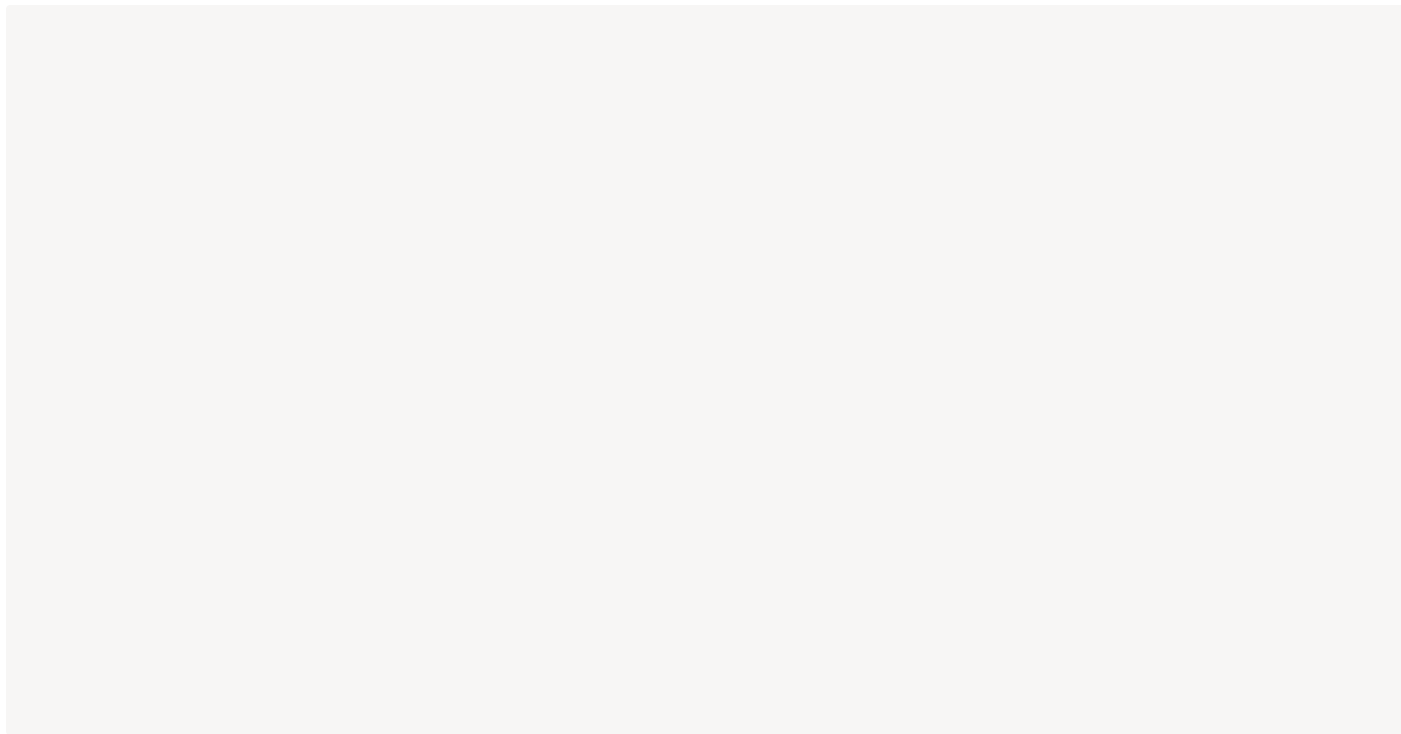


## Creating a dashboard

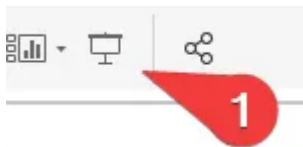
Once we created our first chart, we can add it to the dashboard. To create a new dashboard, click in the *new dashboard* tab



Drag the *sheet1* from the left sidebar



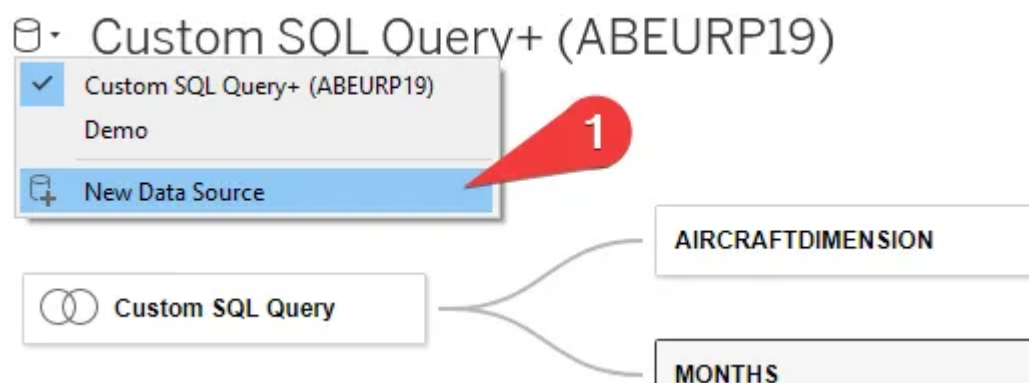
That's it. You can customize it further, but we will see that in a minute. At this stage, you can preview the presentation mode by pressing F7 or clicking in the icon above.



## Adding a second plot to the dashboard

We will now add a chart of ***another data model*** to the dashboard. Consider for example we want to plot **FC/CN, per year**, and we want a dropdown that will let us **filter the data by manufacturer**.

To do so, we cannot rely on the previously created data model. However, Tableau permits us to create multiple data models for satisfying different KPIs in the same dashboard. For this, we will need to create a new data source and add the same target DW database again (see illustration below). You should have now a new *canvas* where to define the new data model.



What tables do you need to answer this query? Try to get to this stage by yourself.

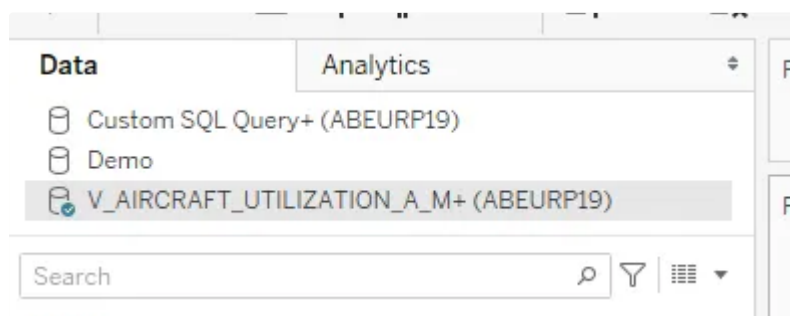


Notice that you don't need to create a new data source for each chart. It can be that several charts can use the same data model from the same data source! However, in some cases to answer some KPIs, you would need to create new data models. For this tutorial, we create a new data model to illustrate this process.

## Custom SQL Query+ (ABEURP19)

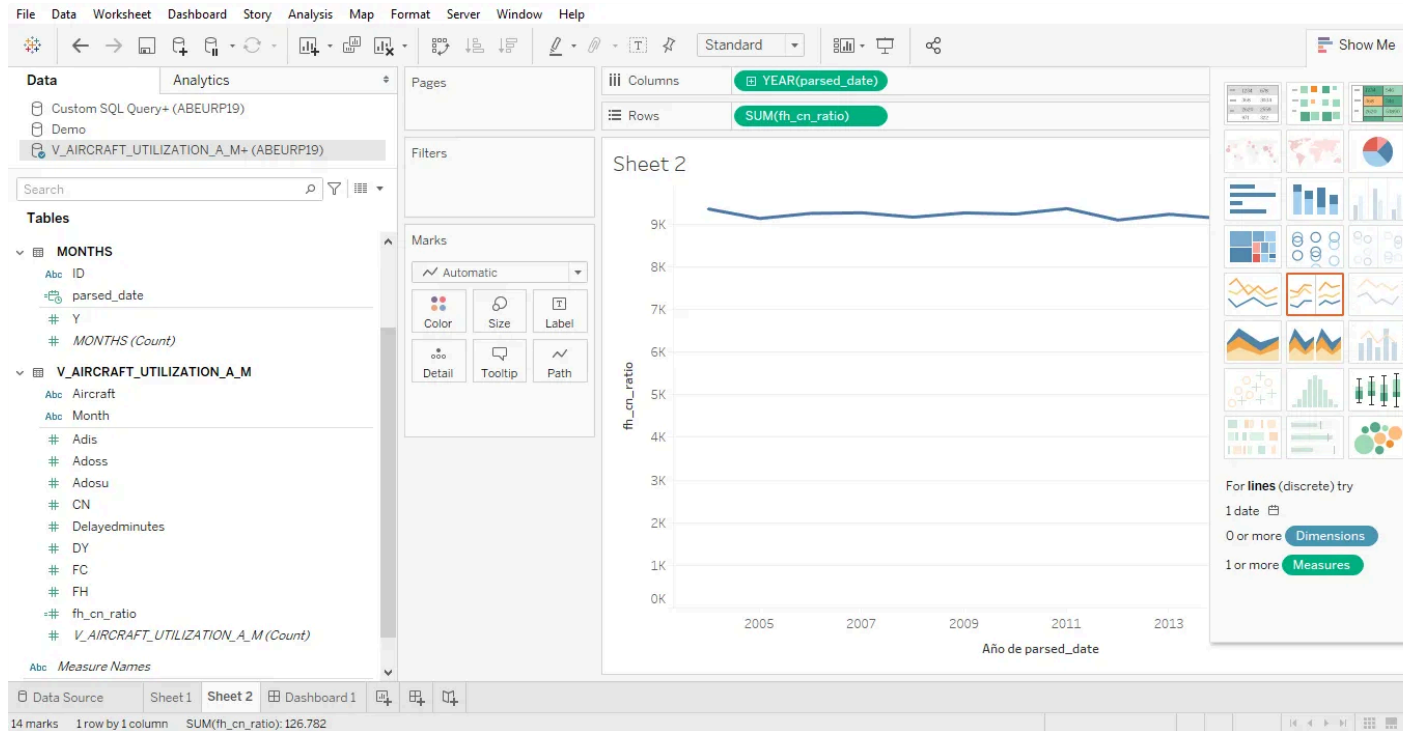


After this, create a *new worksheet* from the tabs below, and repeat the steps again. Check, however, that the data you are using comes indeed from the correct data source.

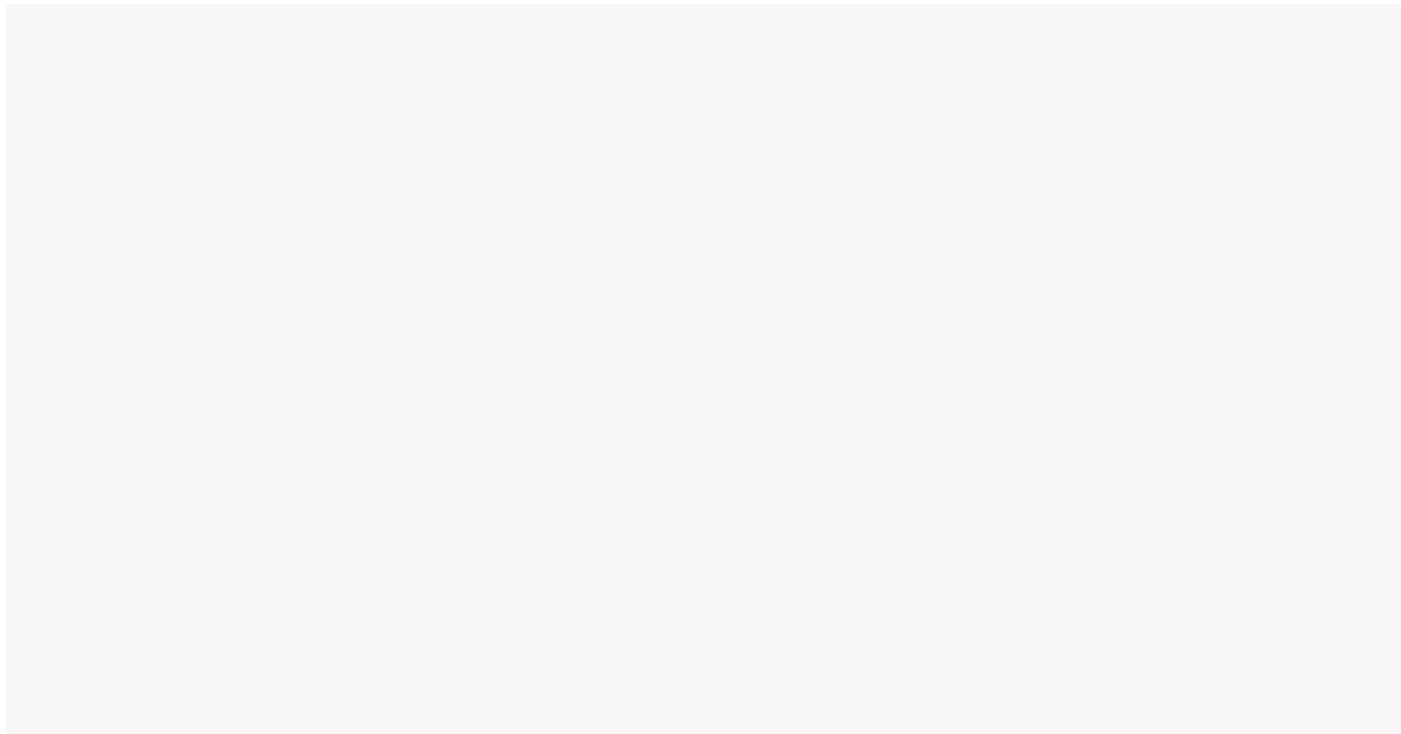


## Calculated fields, again

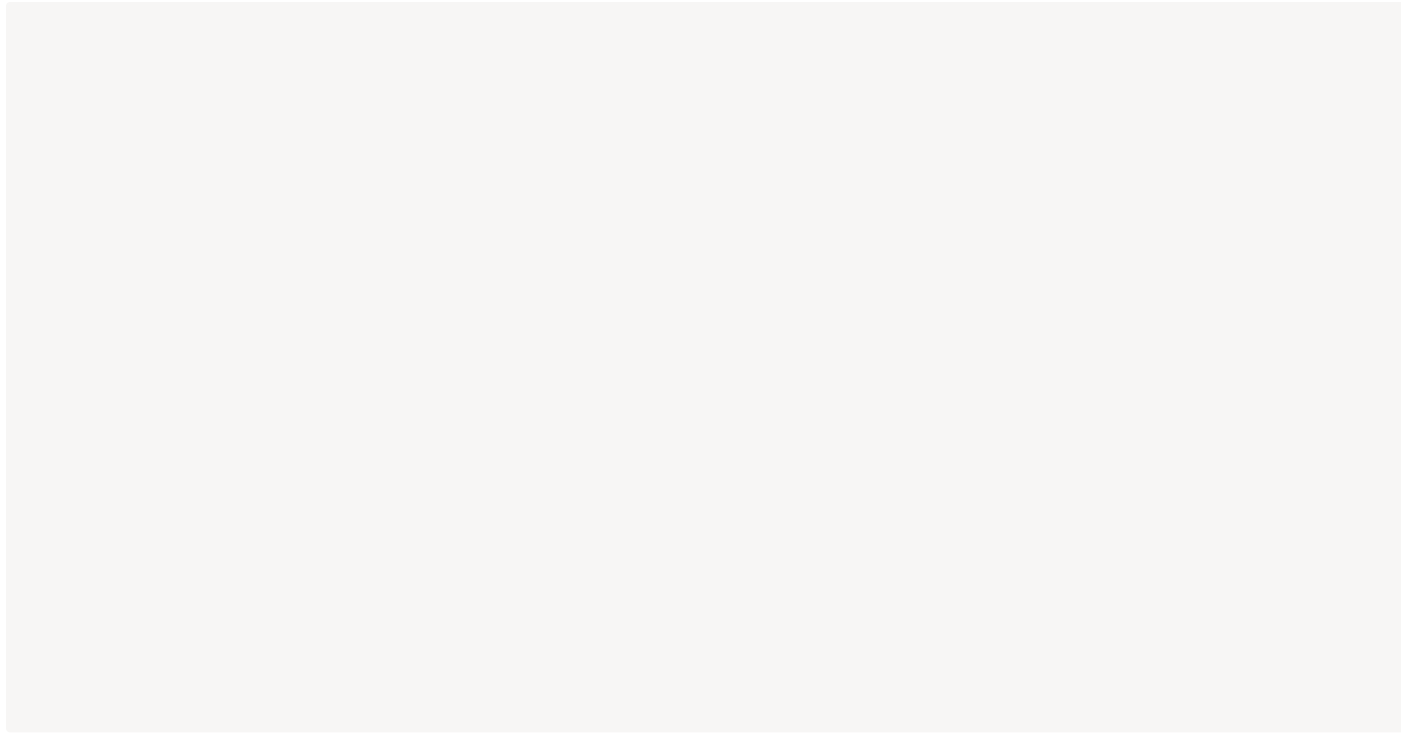
Parse the date from **Months** again (as described before), and also create a new variable called **fh\_cn\_ratio** from **[FH]/[CN]**. You should have something like this by now.



Now, drag the **Manufacturer** from the left sidebar onto the *Pages* area in order to create a dynamic filter (selecting values from the drop down menu). If we use Filters option, we would need to select specific value(s) for which to display the data.



Press F7 now. What happened? Now drag the **Model** into the *Rows* area



We can also display two measures on the same axis in Tableau in order to follow the trends of two measures through time and compare them. For example Add **DY** to the *rows* area, and then on right click choose *Dual axis* option (as shown below).

